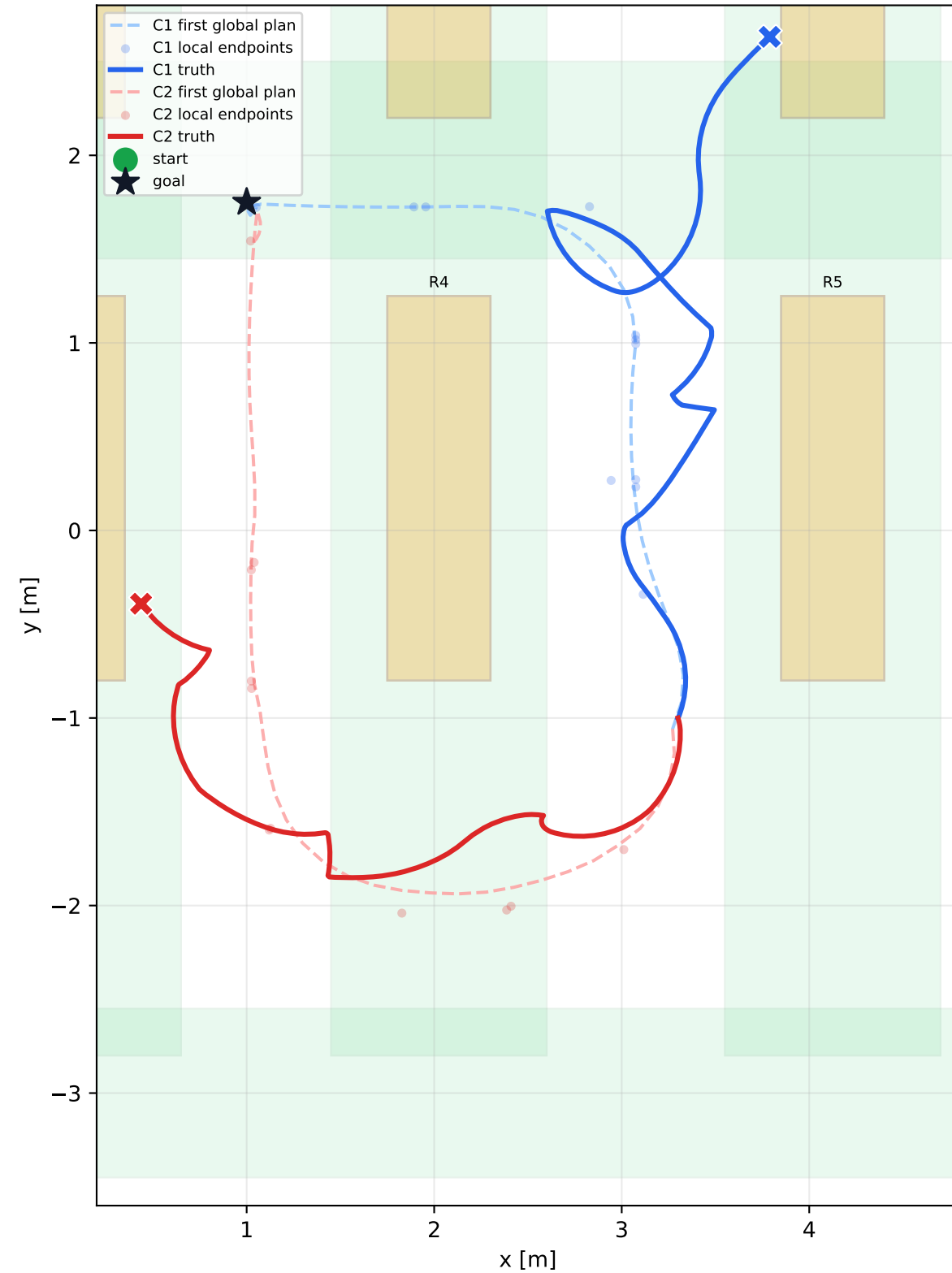


F45 - Tracking yaw + local solver timing diagnostic

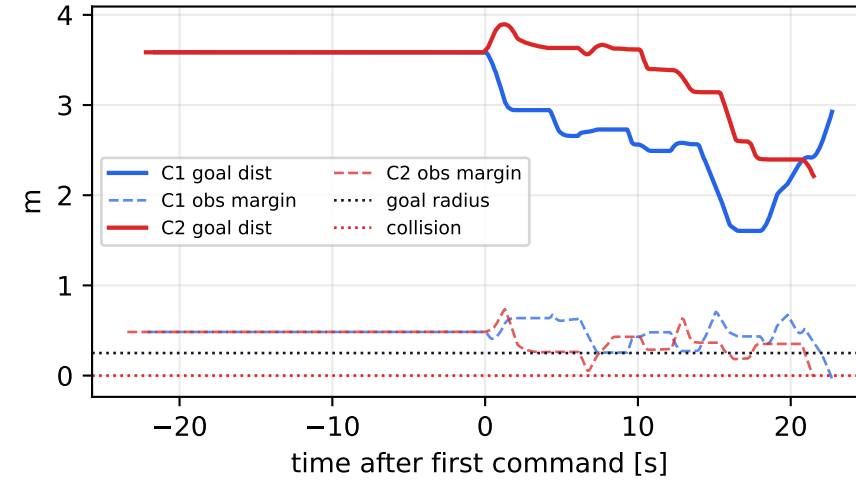
R4

R5

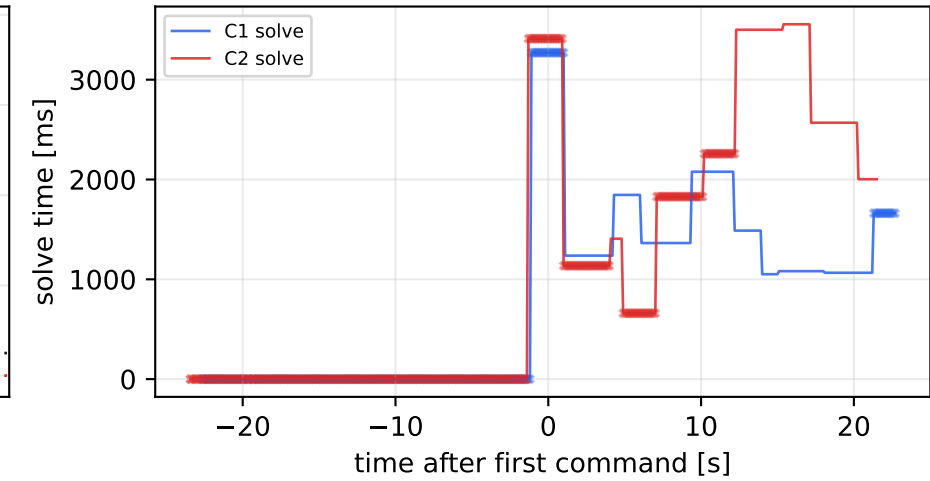
(a) First route, local endpoints, executed trajectory



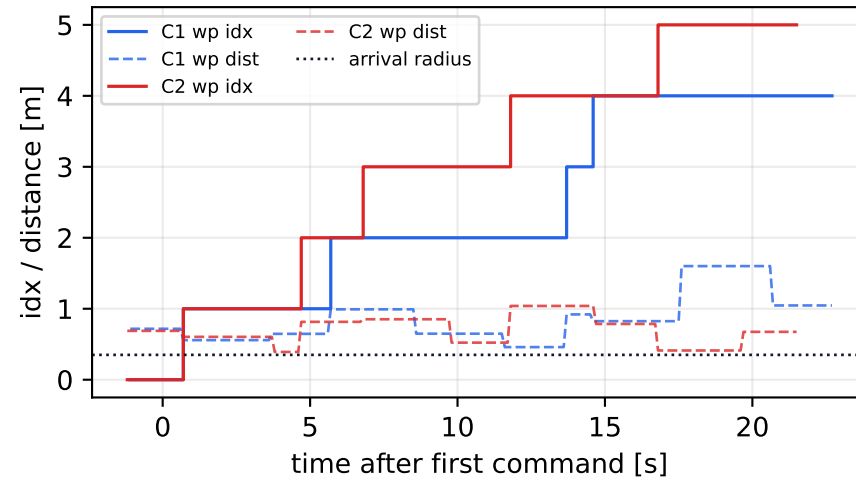
(b) Goal progress and obstacle margin



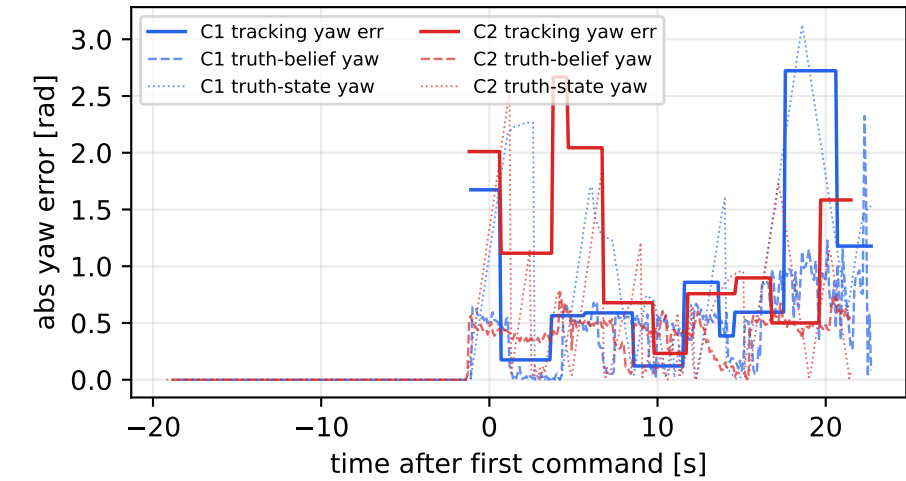
(c) Local solve time (x = non-success)



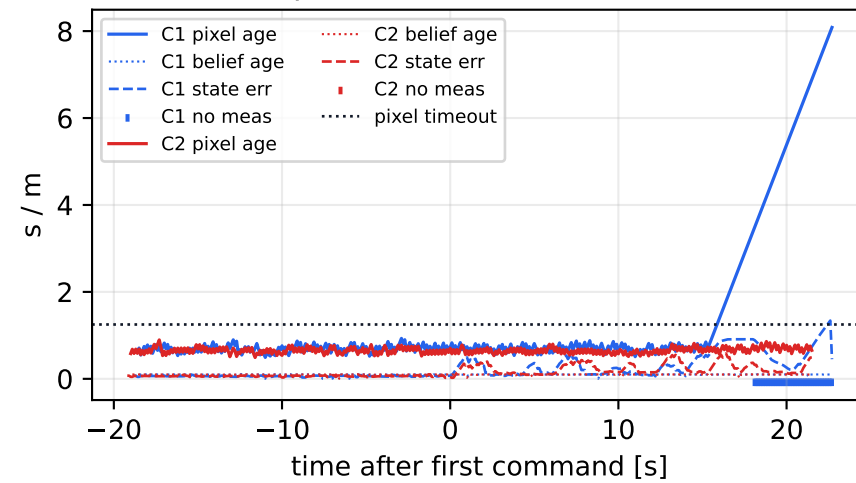
(d) Waypoint progress



(e) Heading/yaw errors



(f) Perception freshness and localization



F45 controlled tracking-yaw smoke

C1: collision, path=6.01m, min_goal=1.60m
err=0.238m, min_obs=-0.039m
solve μ /p90=1614/2197 ms, 46 ms/eval
maxiter/non-success rows=8%
yaw track μ /p90=0.90/2.72 rad
pixel age p90/max=3.95/8.08s

C2: collision, path=4.71m, min_goal=2.21m
err=0.158m, min_obs=-0.013m
solve μ /p90=2233/3506 ms, 76 ms/eval
maxiter/non-success rows=28%
yaw track μ /p90=1.10/2.04 rad
pixel age p90/max=0.75/0.90s

Interpretation:

1. Spawn yaw is clean; yaw problems are runtime/tracking, not init.
2. Global plans still choose the expected route families.
3. Local H15/maxiter20 is too budget-starved: frequent maxiter hits.
4. C1 additionally suffers stale camera belief before collision.
5. C2 localizes better but still collides: local tracking/clearance remains failing.